Listeria Meningitis (MONDO:0006836): A Comprehensive Disease Characterization

Disease: Listeria meningitis / Neurolisteriosis · **MONDO:** MONDO:0006836 · **MeSH:** D008587 (Meningitis, Listeria) · **ICD-10:** A32.11 · **ICD-11:** 1C1A.0 · **Category:** Infectious disease (bacterial, foodborne, opportunistic CNS infection)

Summary

Listeria meningitis is an acute, severe bacterial meningitis/meningoencephalitis caused by the Gram-positive, foodborne, facultative intracellular bacterium *Listeria monocytogenes*. Unlike the classic community-acquired meningitis pathogens (*Streptococcus pneumoniae*, *Neisseria meningitidis*), Listeria is an **"atypical" pathogen** whose disease is almost entirely confined to hosts with impaired cell-mediated immunity: neonates, adults ≥65 years, pregnant women, and people who are diabetic, immunosuppressed, alcoholic, or have malignancy. Because host defense against an intracellular organism depends on T-cell–mediated immunity, the epidemiology is defined by *who cannot mount that response* rather than by exposure alone. Invasive listeriosis is rare (US incidence ≈0.29 per 100,000 per year) but carries a case-fatality of roughly 20–30%, the highest of the common bacterial meningitis etiologies.

Mechanistically, *L. monocytogenes* is a paradigm of intracellular pathogenesis. It crosses the intestinal, placental, and blood–brain barriers using the surface internalins **InlA** and **InlB**, which engage host **E-cadherin (CDH1)** and the **Met** receptor tyrosine kinase, respectively. Once intracellular, the pore-forming toxin **listeriolysin O (LLO/*hly*)** mediates escape from the phagosome into the cytosol, and the surface protein **ActA** nucleates host actin to drive actin-based motility and direct cell-to-cell spread — allowing the bacterium to move between cells while evading humoral immunity. The clinical result is a meningoencephalitis with a characteristic predilection for the **brainstem (rhombencephalitis)**, a pattern that is radiologically and pathologically distinctive and prognostically ominous. Bacterial genotype matters: the hypervirulent **sequence type 6 (ST6/CC6)** independently predicts unfavorable outcome.

Treatment is distinct from empirical meningitis therapy: *L. monocytogenes* is **intrinsically resistant to all cephalosporins**, so first-line therapy is high-dose **ampicillin/amoxicillin, usually combined with gentamicin** for synergy, with **trimethoprim-sulfamethoxazole**

(TMP-SMX) as the principal alternative in penicillin-allergic patients. Adjunctive dexamethasone, standard in pneumococcal meningitis, is generally stopped once Listeria is confirmed. No vaccine exists; prevention rests on **food-safety counseling** of high-risk groups (avoiding deli meats, soft cheeses, unpasteurized dairy) and public-health surveillance (whole-genome sequencing of clinical and food-processing isolates). This report synthesizes 10 confirmed findings from 49 reviewed papers across all 15 requested disease-characteristic domains.

Key Findings

Finding 1 — Listeria is an "atypical" meningitis pathogen concentrated in the elderly, diabetic, and immunosuppressed

L. monocytogenes is a minority cause of community-acquired bacterial meningitis overall, but it is strongly over-represented in specific vulnerable populations, reflecting the requirement for intact cell-mediated immunity to control an intracellular pathogen.

In a Dutch nationwide prospective cohort, **diabetes conferred a 2.2-fold higher risk of acquiring bacterial meningitis** and was an independent risk factor for death (adjusted OR 1.98, 95% CI 1.13–3.48, P=0.017); Listeria caused 6% of meningitis episodes in diabetic patients (PMID: 27845429). Among patients **≥80 years**, atypical pathogens including *L. monocytogenes* occurred substantially more often than in younger adults (49/149 [33%] vs 362/1991 [18%]; p<0.001) (PMID: 35352336). A Barcelona cohort likewise found Listeria more frequent among diabetics (18% vs 10%, p=0.033) (PMID: 32312231). Additional risk-group associations include alcoholism (Listeria caused 8% of alcoholic meningitis, with 25% mortality; PMID: 28065808), immunosuppressive medication (chiefly corticosteroids; PMID: 27613024), and systemic lupus erythematosus (PMID: 20213126).

> *"the risk of acquiring bacterial meningitis was 2.2-fold higher for diabetes patients"* — PMID: 27845429

Interpretation: The disease is defined by host immune status. This has direct clinical consequence — any adult ≥50, or any immunocompromised or diabetic patient, presenting with meningitis should receive **empirical ampicillin to cover Listeria** in addition to a cephalosporin/vancomycin regimen.

Finding 2 — The ST6/CC6 genotype independently predicts unfavorable outcome

Outcome is partly determined by the *bacterial* genotype, not only the host. In 92 adults with *L. monocytogenes* meningitis pooled from two Dutch nationwide cohorts, the rate of unfavorable outcome rose from 27% (1998–2002) to 61% (2006–2012, P=0.002). Multivariate regression identified **infection with sequence type 6 (ST6) as the sole independent predictor of unfavorable outcome (OR 3.77, 95% CI 1.07–13.33)**, and ST6 became more prevalent over time (4% vs 29%, P=0.009) (PMID: 23592828).

> *"Multivariate regression analysis identified infection with L. monocytogenes ST6 as the sole predictor of unfavorable outcome (odds ratio, 3.77; 95% confidence interval, 1.07-13.33)"* — PMID: 23592828

This links to the genomic epidemiology of hypervirulent clonal complex 1 (CC1) and related lineages (PMID: 38165396) and is reinforced by animal-model data (Finding 6) showing genotype-dependent virulence.

Finding 3 — Neurolisteriosis presents chiefly as meningoencephalitis, with characteristic brainstem involvement

In the French **MONALISA** national prospective cohort (818 cases: 107 maternal-neonatal, 427 bacteraemia, 252 neurolisteriosis), **neurolisteriosis presented as meningoencephalitis in 212/252 (84%)** and **brainstem (rhombencephalitis) involvement in 42/252 (17%)**. Three-month mortality was higher for pure bacteraemia than neurolisteriosis (HR 0.54, 95% CI 0.41–0.69, p<0.0001). Among pregnant women, 26/107 (24%) experienced fetal loss — notably, never after 29 weeks' gestation (PMID: 28139432).

> *"Neurolisteriosis presented as meningoencephalitis in 212 (84%) of 252 patients; brainstem involvement was only reported in 42 (17%) of 252 patients"* — PMID: 28139432

Historically, in an epidemic Swiss series, 6 of 8 adults with parenchymal CNS infection had brainstem involvement, and a preceding influenza-like prodrome was common (PMID: 3918344). The predilection for the brainstem distinguishes Listeria clinically from other bacterial meningitides.

Finding 4 — CNS invasion is mediated by internalins (InlA/InlB → E-cadherin/Met) with LLO/ActA driving intracellular survival and spread

The molecular pathogenesis is well characterized. **Internalins InlA and InlB interact with host receptors E-cadherin and MET (mesenchymal-epithelial transition factor) to facilitate brain invasion**, and *L. monocytogenes* activates NF-κB in brain endothelium (PMID: 30117097). Once internalized, **listeriolysin O (LLO)** mediates phagosomal escape into the cytosol, and **ActA** drives actin-based motility and cell-to-cell spread. In the zebrafish larval model, bacteria escaped the macrophage phagosome in an LLO-dependent manner, an event critical for virulence (PMID: 19546195); LLO additionally promotes protrusion formation and spread by exploiting **efferocytosis** (PMID: 24739967).

> *"Internalins (InlA and InlB) of Listeria monocytogenes interact with its cellular receptors E-cadherin and mesenchymal-epithelial transition (MET) to facilitate invading the brain"* — PMID: 30117097
>
> *"bacteria could escape the macrophage phagosome in a listeriolysin-dependent manner and accessed the cytosol; this event was critical for bacterial virulence"* — PMID: 19546195

Secretion of these virulence factors depends on the **Sec/SecDF** translocation system (PMID: 24056100) and is governed by the master transcriptional regulator **PrfA** (PMID: 26055558). Passive immunization with anti-ActA and anti-LLO antibodies is protective in mice, confirming these factors as functional virulence determinants (PMID: 28004800).

Finding 5 — Ampicillin/amoxicillin (± gentamicin) is first-line; cephalosporins fail; TMP-SMX is the key alternative

L. monocytogenes is **intrinsically resistant to cephalosporins**, the single most important therapeutic fact about this disease. Guidelines specify amoxicillin (+gentamicin) for Listeria and continue **amoxicillin alone for 14 days** in confirmed Listeria meningitis; cefotaxime is reserved for *E. coli* precisely because Listeria is not covered by cephalosporins (PMID: 29433694). In an infant-rat meningoencephalitis model, **amoxicillin plus gentamicin was the**

most active regimen in brain tissue (PMID: 10390217). **TMP-SMX** has bactericidal intracellular activity and excellent CNS penetration and has cured refractory listeriosis unresponsive to amoxicillin/gentamicin (PMID: 10959905).

> *"In case of meningitis due to GBS or L. monocytogenes antibiotherapy is continued for 14 days with amoxicillin alone"* — PMID: 29433694
>
> *"TMP/SMX has bactericidal extracellular and intracellular activity against Listeria and excellent central nervous system penetration, and thus may be effective for the treatment of refractory listeriosis"* — PMID: 10959905

Regarding **adjunctive dexamethasone**: while corticosteroids reduce mortality and hearing loss in bacterial meningitis overall (largely driven by pneumococcus; PMID: 17253505, PMID: 23152589), current guidance is to **stop dexamethasone once Listeria is confirmed** (PMID: 36472590, PMID: 42601017).

Finding 6 — Animal models recapitulate the disease and reveal genotype and species specificity

A validated **C57BL/6 intracisternal *L. monocytogenes* meningitis mouse model** demonstrated genotype-dependent severity: **ST1-infected mice had a more severe inflammatory response and brain damage than ST6-infected mice** (PMID: 30193592). Because the **InlA:E-cadherin interaction is species-specific** (human vs murine E-cadherin differ at a critical residue), **transgenic mice expressing human E-cadherin** were generated, in which the essential role of InlA in crossing the intestinal barrier was established (PMID: 15060982). The **zebrafish larval model** reproduces LLO-dependent phagosomal escape and ActA-dependent spread with real-time imaging (PMID: 19546195).

> *"ST1-infected mice had a more severe inflammatory response and brain damage as compared to ST6-infected mice"* — PMID: 30193592

Finding 7 — Invasive listeriosis is rare but severe, concentrated in the elderly and pregnant

US CDC surveillance (2009–2011, 1,651 cases) reported an **average annual incidence of 0.29 cases/100,000**, a **case-fatality rate of 21%**, with **58% of cases in adults ≥65 years** and **14% pregnancy-associated**. Incidence was markedly higher among adults ≥65 (1.3/100,000; **relative rate 4.4**) and pregnant women (3.0/100,000; **RR 10.1**); ≥74% of non-pregnant patients <65 had an immunocompromising condition (PMID: 23739339). A global 80-year meta-analysis found **Listeria meningitis had the highest case-fatality (27%, 95% CI 24–31%)** of the common causes, exceeding pneumococcus (24%) and far exceeding meningococcus (9%) (PMID: 39093565).

> *"The average annual incidence was 0.29 cases per 100,000 population. Compared with the overall population, incidence was markedly higher among adults aged ≥65 years (1.3; relative rate [RR]: 4.4) and pregnant women (3.0; RR: 10.1)"* — PMID: 23739339

Finding 8 — Neuroimaging shows rhombencephalitis, abscesses, and hydrocephalus with prognostic weight

In the MONALISA prospective imaging study (71 neurolisteriosis patients, mean age 64), 85% had encephalitis signs and imaging was abnormal in 87%: meningeal enhancement (35%), abscess/nodules (14%), hemorrhages (15%), hydrocephalus/contrast-enhancing ventricles (10%). **Three-month survival was lower with hydrocephalus (14% vs 73%, P=0.005) and with parenchymal lesions (54% vs 92%, P=0.004)** (PMID: 29796652). A Granada cohort found rhombencephalitis in 10/120 (8.3%), with MRI showing **T2-FLAIR hyperintensity (100%)**, pons/medulla/cerebellum involvement, and cranial-nerve enhancement (70%); rhombencephalitis increased in-hospital mortality (PMID: 37303100). The classic MRI appearance of Listeria rhombencephalitis has been described since the 1990s (PMID: 8826752).

> *"The most frequent MRI findings in patients with confirmed rhombencephalitis were T2-FLAIR hyperintensity (100%)... while the most frequent anatomical involvement were pons, medulla oblongata, and cerebellum"* — PMID: 37303100

Finding 9 — Neurolisteriosis is a shared zoonotic syndrome ("circling disease") in ruminants

Bovine listeriosis mirrors the human disease: 11/12 cases presented as neurolisteriosis with **suppurative meningoencephalitis and brainstem microabscesses**, and isolates were serotypes 4b and 1/2a (shared with humans) carrying the *inlA* virulence gene (PMID: 40192846). In goats with encephalitic listeriosis, circling (or a history of circling) was associated with survival (OR 6.24), and males were 14× more likely to be non-survivors (PMID: 37084183). Ruminant isolates display zoonotic characteristics, causing similar clinical signs in humans including abortion and encephalitis (PMID: 42667904).

> *"its ruminant isolates display zoonotic characteristics, causing similar clinical signs in humans, including abortion and encephalitis"* — PMID: 42667904

Finding 10 — Neonatal listeriosis is bimodal: early-onset (in-utero sepsis) vs late-onset (meningitis)

In a Swedish series (1958–1974, 110 cases), of 37 liveborn neonates, **22 had "early disease" (mostly granulomatosis infantiseptica)** acquired in utero from the mother, and **8 had "late disease" with meningoencephalitis** from often nosocomial postnatal sources; maternal cases included 8 abortions and 3 stillbirths, with serotypes 1 and 4b predominating (PMID: 419368). Modern reviews confirm foodborne maternal acquisition (deli meats, unprocessed cheese) placing neonates at high morbidity risk (PMID: 40887071).

> *"Out of 37 neonates born alive, 22 had 'early disease', mostly granulomatosis infantiseptica, and 8 had 'late disease' with meningoencephalitis"* — PMID: 419368

Section-by-Section Disease Characterization

1. Disease Information

Listeria meningitis is an acute bacterial infection of the meninges (and frequently the brain parenchyma, hence meningoencephalitis) caused by *Listeria monocytogenes*. The data derive from **aggregated disease-level resources** — national prospective cohorts (Dutch MeningItis cohorts, French MONALISA), CDC surveillance, and case series — rather than a single-gene EHR phenotype; this is an infectious rather than a Mendelian entity.

Identifier system	Value
MONDO	MONDO:0006836 (Listeria meningitis)
MeSH	*Meningitis, Listeria* (D008587); *Listeriosis* (D008088)
ICD-10	A32.11 (Listerial meningitis); A32.12 (Listerial meningoencephalitis); P37.2 (neonatal)
ICD-11	1C1A.0 (Listeriosis, with CNS extension)
NCBI Taxonomy	*Listeria monocytogenes* — txid1639

Synonyms: listerial meningitis, neurolisteriosis (broader term including meningoencephalitis, rhombencephalitis, brain abscess), CNS listeriosis. Not an OMIM/Orphanet Mendelian disease.

2. Etiology

Primary cause: infection with *L. monocytogenes*, a foodborne facultative intracellular Gram-positive bacillus (Findings 1, 4, 5). The route is ingestion of contaminated food (deli meats, soft/unpasteurized cheeses, unprocessed dairy, produce) → intestinal translocation → bacteremia → CNS/placental seeding.

Environmental/host risk factors: age ≥65 (RR 4.4), pregnancy (RR 10.1), neonatal status, diabetes mellitus (2.2× meningitis risk), immunosuppressive medication (chiefly corticosteroids), alcoholism, malignancy, SLE, and other conditions impairing cell-mediated immunity (Findings 1, 7; PMIDs 27845429, 35352336, 32312231, 28065808, 27613024, 20213126, 23739339).

Genetic risk factors (host): No Mendelian causal genes. Susceptibility is driven by acquired/iatrogenic immune compromise. No established GWAS locus specific to Listeria meningitis was found — a **knowledge gap** — though defects in cell-mediated immunity (e.g., TNF-α blockade, corticosteroids) are mechanistically implicated.

Pathogen genetic factors: virulence and outcome vary by bacterial genotype — hypervirulent clonal complexes CC1, CC4, CC6; **ST6 independently predicts poor outcome** (Finding 2; P 23592828).

Key virulence genes: *inlA*, *inlB*, *hly* (LLO), *actA*, *plcA/plcB*, regulated by *prfA*.

Protective factors: intact cell-mediated (T-cell) immunity is the principal protective factor. Behaviorally, avoidance of high-risk foods reduces exposure. No genetic protective variant is established.

3. Phenotypes

Phenotype	Type	Frequency / notes	Suggested HPO
Fever	Symptom	~74–75%	HP:0001945
Headache	Symptom	Common (~84% in general BM)	HP:0002315
Neck stiffness / meningismus	Clinical sign	~74%	HP:0031179 / HP:0000708
Altered mental status / drowsiness	Sign	Very common; part of classic triad	HP:0011446 / HP:0002329
Meningoencephalitis	Syndrome	84% of neurolisteriosis	HP:0002383
Rhombencephalitis / brainstem signs (ataxia, cranial neuropathies, nystagmus, dysphagia)	Sign	~8–17%	HP:0002401, HP:0001251, HP:0000618
Seizures	Sign	Elevated in alcoholic/severe cases	HP:0001250
CSF pleocytosis, low glucose, high protein	Lab abnormality	Diagnostic	HP:0002921
Hydrocephalus	Complication	~10%; poor prognosis	HP:0000238
Fetal loss / stillbirth (maternal)	Outcome	24% in pregnant	HP:0001561, HP:0003826

Onset is **acute to subacute**; a preceding influenza-like prodrome is characteristic (PMID: 3918344). Severity is high (case-fatality 20–30%). Quality-of-life impact among survivors of bacterial meningitis broadly is substantial: at 12 months, 26.7% had hearing loss, 32.8% depressive symptoms, 31.0% persistent headache (PMID: 31927117).

4. Genetic/Molecular Information

Not applicable for host germline genetics — Listeria meningitis is infectious, with no causal human gene, pathogenic variant, ACMG classification, or inheritance pattern. The relevant "genetics" is the **bacterial genome**: virulence loci *inlA*/*inlB* (internalins → E-cadherin/Met), *hly* (listeriolysin O), *actA* (actin-based motility), *plcA*/*plcB* (phospholipases), master regulator *prfA*. Genotype (ST/CC) modifies outcome (Finding 2). Host HGNC genes engaged as receptors: **CDH1** (E-cadherin) and **MET**.

5. Environmental Information

Infectious agent: *Listeria monocytogenes* (NCBI txid1639); serotypes 4b, 1/2a, 1/2b are most clinically important. **Environmental reservoir:** soil, water, decaying vegetation; contaminates food-processing environments via biofilm formation (PMID: 38287241). **Lifestyle/dietary factors:** consumption of deli meats, pâté, soft cheeses, unpasteurized dairy, and refrigerated ready-to-eat foods (Listeria grows at refrigeration temperatures). Alcohol use is a host risk factor (PMID: 28065808).

6. Mechanism / Pathophysiology — Causal Chain

```
1. Ingestion of L. monocytogenes-contaminated food
        | leads to
2. InlA binds host E-cadherin (CDH1) on intestinal epithelium → invasion & translocation across
   InlB binds Met receptor → complementary invasion of non-epithelial cells
        | results in
3. Bacteremia (facultative intracellular survival in monocytes/macrophages)
        | branches:
   3a → Placental barrier crossing (InlA/InlB) → fetal/neonatal infection → granulomatosis infar
   3b → Blood-brain barrier crossing (endothelial invasion; NF-κB activation; possibly "Trojan-H
        | leads to
4. CNS invasion → within cells, LLO (hly) perforates the phagosome → cytosolic escape (critical
        | results in
5. ActA nucleates host actin → actin-based motility → cell-to-cell spread (evades humoral immuni
        | leads to
6. Neutrophilic/suppurative inflammation of meninges and brain parenchyma; predilection for brai
        | results in
7. Meningoencephalitis / rhombencephalitis, abscesses, hydrocephalus, hemorrhage
        | leads to
8. Raised ICP, cranial neuropathies, coma → death (case-fatality ~20-30%) or neurological sequel
```

Steps 2, 4, 5 are **experimentally demonstrated** (PMIDs 30117097, 19546195, 24739967, 15060982). Step 3b's precise BBB-crossing route (transcellular vs Trojan-horse) is partly **inferred**. **Molecular pathways:** host **NF-κB** activation in endothelium (P 30117097); PrfA-regulated virulence regulon in the bacterium. **Cellular processes (GO):** GO:0006909 phagocytosis, GO:0090662 (phagosome), GO:0006935 actin-based motility, GO:0043277 (apoptotic cell clearance/efferocytosis), GO:0006954 inflammatory response. **Cell types (CL):** macrophages (CL:0000235), monocytes (CL:0000576), neutrophils (CL:0000775), brain microvascular endothelial cells (CL:0002139). Neutrophils can provide an intravacuolar niche (PMID: 40067021).

7. Anatomical Structures Affected

- **Primary organ:** brain and meninges — leptomeninges (UBERON:0002361), brain (UBERON:0000955), with characteristic **brainstem** (UBERON:0002298), **pons** (UBERON:0000988), **medulla oblongata** (UBERON:0001896), and **cerebellum** (UBERON:0002037) involvement (Findings 3, 8).
- **Body system:** central nervous system.
- **Secondary:** bloodstream (bacteremia/sepsis), placenta (UBERON:0001987) and fetus in pregnancy; ventricles (hydrocephalus).
- **Tissue/cell level:** nervous tissue; meningeal/epithelial and endothelial barriers; targets macrophages, monocytes, neutrophils, endothelial cells.
- **Subcellular (GO CC):** phagosome (GO:0045335), cytosol (GO:0005829), host actin cytoskeleton (GO:0015629); host plasma membrane (E-cadherin/Met receptors).
- **Lateralization:** brainstem lesions often bilateral/asymmetric; meningitis diffuse.

8. Temporal Development

Onset is **acute to subacute** (days), often after an influenza-like prodrome. Age of onset is bimodal at the population level: **neonatal** (early-onset in-utero sepsis vs late-onset meningitis; Finding 10) and **elderly/immunocompromised adult**. Course without treatment is rapidly progressive to coma and death; with treatment, disease is **self-limited over a 2–3 week antibiotic course** (14 days minimum for meningitis, longer for rhombencephalitis/abscess). Complications (hydrocephalus, parenchymal lesions) mark a worse trajectory (Finding 8). The critical intervention window is prompt empirical ampicillin — delay worsens outcome.

9. Inheritance and Population

- **Inheritance:** Not applicable (infectious).
- **Epidemiology:** US incidence ≈0.29/100,000/year overall; 1.3/100,000 in ≥65 (RR 4.4); 3.0/100,000 in pregnancy (RR 10.1) (Finding 7). 58% of cases in ≥65; 14% pregnancy-associated.
- **Sex:** Several series show female predominance driven by pregnancy-associated cases (e.g., 69.8% female in a Karachi cohort, PMID: 42018538; 77% female in Qatar, PMID: 40978187); outside pregnancy, elderly males predominate.
- **Geographic distribution:** worldwide; outbreaks tied to contaminated food products; winter seasonal peaks reported in some series (PMID: 3918344).

10. Diagnostics

- **Definitive:** CSF and/or blood **culture** of *L. monocytogenes* (Gram-positive rods; may be sparse). Gram-stain sensitivity is limited.
- **CSF profile:** pleocytosis (may be modest/mononuclear, especially in immunosuppressed — <1000/mm³ in ~46% of immunosuppressed; PMID: 27613024), elevated protein, low glucose.
- **Molecular:** PCR / multiplex meningitis-encephalitis panels; whole-genome sequencing for genotyping/surveillance (Findings 2, 10;

P 40887071

).
- **Imaging:** MRI is key — T2-FLAIR hyperintensity of pons/medulla/cerebellum, meningeal enhancement, abscesses, hydrocephalus, hemorrhage (Finding 8; PMIDs 29796652, 37303100, 8826752).
- **Differential diagnosis:** other bacterial meningitides (pneumococcus, meningococcus), viral encephalitis (esp. HSV rhombencephalitis), tuberculous meningitis, brain abscess, and — in SLE patients — lupus flare (which Listeria can mimic; PMID: 20213126).

11. Outcome / Prognosis

- **Case-fatality:** ~20–30% (US surveillance 21%; global meta-analysis 27%, the highest of common causes) (Finding 7; PMIDs 23739339, 39093565).
- **Prognostic factors:** older age; bacterial genotype **ST6** (OR 3.77; Finding 2); imaging findings — **hydrocephalus** (survival 14% vs 73%) and parenchymal lesions (54% vs 92%) (Finding 8); coma/altered consciousness; intracranial hemorrhage (associated with 63% mortality; PMID: 23028898).

- **Sequelae:** hearing loss, persistent neurological deficits, cognitive/depressive symptoms among survivors (PMID: 31927117).
- **Maternal-fetal:** high fetal loss (24%), stillbirth, preterm delivery; mothers often recover fully.

12. Treatment

Intervention	Role	NCIT (suggested)
Ampicillin / amoxicillin (high-dose IV)	First-line; 14+ days for meningitis	NCIT:C356 / C61785
Gentamicin (added for synergy)	Adjunct, esp. early/severe	NCIT:C544
TMP-SMX (co-trimoxazole)	Alternative (penicillin allergy); CNS-penetrant, intracellular; salvage for refractory disease	NCIT:C312
Meropenem	Alternative in some settings	NCIT:C61796
Cephalosporins	**Ineffective — intrinsic resistance** (do not use)	—
Adjunctive **dexamethasone**	Standard empirically, **stop when Listeria confirmed**	NCIT:C583
Supportive / neurocritical care	ICP management, seizure control, CSF diversion for hydrocephalus	—

Evidence: Finding 5 (PMIDs 29433694, 10390217, 10959905); dexamethasone context (PMIDs 17253505, 23152589, 42601017, 36472590). No pharmacogenomic, gene, cell, or RNA-based therapy applies. Passive immunization (anti-ActA/anti-LLO) is experimental/preclinical only (PMID: 28004800).

13. Prevention

- **Primary (no vaccine exists):** dietary/behavioral counseling of high-risk groups — pregnant women, elderly, immunocompromised — to avoid deli meats, pâté, soft/unpasteurized cheeses, unpasteurized milk, and to reheat ready-to-eat foods; refrigeration hygiene (PMID: 40887071, PMID: 42018538).

- **Public health:** food-safety regulation, environmental monitoring of processing plants, and **whole-genome-sequencing surveillance** linking clinical and food isolates to detect and interrupt outbreaks
  (P 40887071 ; PMID: 11938118).
- **Secondary/tertiary:** prompt empirical ampicillin in at-risk meningitis patients; neurocritical monitoring to prevent complications.
- No genetic screening, immunization, or chemoprophylaxis is applicable.

14. Other Species / Natural Disease

Listeria neurolisteriosis is a well-recognized **zoonotic** and veterinary disease. In ruminants (cattle, sheep, goats) it causes **"circling disease"** — suppurative meningoencephalitis with **brainstem microabscesses** that pathologically parallel human rhombencephalitis (Finding 9; PMIDs 40192846, 37084183, 42667904). Shared serotypes (4b, 1/2a) and virulence genes (*inlA*) underscore cross-species conservation of mechanism. Species affected: *Bos taurus* (txid9913), *Capra hircus* (txid9925), *Ovis aries* (txid9940). Orthologous host receptor genes (CDH1, MET) are conserved, though InlA:E-cadherin binding is species-specific (Finding 6). Ovine-derived isolates infect trigeminal ganglion neurons via InlA/InlB/LLO (PMID: 42121802), a proposed route of neuroinvasion in ruminants.

15. Model Organisms

Model	Type	Recapitulation	PMID
C57BL/6 intracisternal meningitis mouse	Mammalian, induced	Meningitis with genotype-dependent severity (ST1>ST6)	30193592
Humanized E-cadherin transgenic mouse	Mammalian, genetic (knock-in)	Overcomes InlA species specificity; models intestinal barrier crossing	15060982
Oral infection murine model (ruminant isolates)	Mammalian, induced	Pathogenicity/genomic comparison	42667904
Infant-rat meningoencephalitis	Mammalian, induced	Antibiotic efficacy testing in brain	10390217
Zebrafish larvae	Vertebrate, in vivo imaging	LLO-dependent phagosomal escape; ActA-dependent spread; real-time phagocyte interaction	19546195
RAW264.7 / cultured cell lines	In vitro	Invasion, intracellular growth, cell-to-cell spread	28004800, 26055558

Limitations: wild-type mice under-model oral/intestinal invasion because murine E-cadherin does not bind InlA (requires humanized model); no single model fully captures the elderly/immunocompromised human host context.

Mechanistic Model / Interpretation

The unifying theme is that **Listeria meningitis is a disease of the intersection between an intracellular pathogen's molecular toolkit and a host who cannot deploy cell-mediated immunity.** Three axes explain nearly all clinical observations:

1. **Host immune axis** (Findings 1, 7) — defines *who* gets the disease: neonates, elderly, pregnant, diabetic, immunosuppressed. This drives the epidemiology (RR 4.4–10.1) and the demand for empirical ampicillin in at-risk meningitis.

2. **Pathogen invasion/spread axis** (Findings 4, 6) — defines *how* the disease happens: InlA/InlB→E-cadherin/Met for barrier crossing (gut, placenta, BBB), LLO for phagosomal escape, ActA for intracellular spread. This intracellular, cell-to-cell lifestyle explains why humoral immunity and cephalosporins fail while cell-penetrant agents (ampicillin, TMP-SMX) succeed.
3. **Pathogen genotype axis** (Findings 2, 6) — defines *how badly*: hypervirulent ST6/CC lineages independently worsen outcome, corroborated by genotype-dependent murine brain damage.

These converge on a **brainstem-predominant meningoencephalitis** (Findings 3, 8) whose imaging signature (T2-FLAIR pons/medulla/cerebellum, hydrocephalus) is both diagnostic and prognostic, and which is faithfully mirrored in ruminant "circling disease" (Finding 9), confirming a deeply conserved zoonotic syndrome.

Evidence Base

PMID	Contribution	Supports
27845429	Diabetes 2.2× BM risk; Listeria 6%	F1
35352336	Atypical pathogens 33% in ≥80y	F1
23592828	ST6 sole predictor of poor outcome	F2
28139432	MONALISA: 84% meningoencephalitis, 17% brainstem	F3
30117097	InlA/InlB→E-cadherin/Met brain invasion	F4
19546195	LLO-dependent escape; zebrafish model	F4, F6
24739967	LLO/efferocytosis-mediated spread	F4
29433694	Amoxicillin 14d; cephalosporins fail	F5
10959905	TMP-SMX CNS-penetrant salvage	F5
10390217	Amox+gent most active in brain	F5
30193592	Mouse meningitis model, ST1>ST6	F6
15060982	Humanized E-cadherin mouse; InlA specificity	F6
23739339	US incidence 0.29, CFR 21%, RR by group	F7
39093565	Listeria highest CFR (27%) among causes	F7
29796652	Imaging; hydrocephalus/parenchymal prognosis	F8
37303100	Rhombencephalitis MRI signature	F8
40192846	Bovine brainstem microabscesses	F9
42667904	Zoonotic ruminant isolates	F9
419368	Bimodal neonatal disease	F10
40887071	Neonatal listeriosis review/prevention	F10

Limitations and Knowledge Gaps

- **Host genetic susceptibility is undefined.** No GWAS locus or Mendelian modifier for Listeria meningitis specifically was identified; susceptibility is framed entirely through acquired immune compromise. Section 4 (host genetics) and much of Section 9 (inheritance) are **not applicable** to this infectious disease.
- **BBB-crossing route is partially inferred.** Whether Listeria crosses the BBB primarily transcellularly, paracellularly, or via infected leukocytes ("Trojan horse") is not fully resolved in the reviewed evidence.
- **Sex-ratio data are confounded** by the large pregnancy-associated fraction; true non-pregnant sex ratios are less well quantified.
- **Metabolomics/proteomics/single-cell/spatial** profiling of human Listeria meningitis was not found — these omics domains remain gaps.
- **Prognostic value of ST6** derives largely from Dutch cohorts; generalizability across geographies needs confirmation.
- **No randomized trial** defines the optimal antibiotic regimen or the role of gentamicin/dexamethasone specifically in Listeria meningitis; guidance is extrapolated from cohorts and models.

Proposed Follow-up Experiments / Actions

1. **Genotype–outcome validation:** Prospective multinational cohort correlating *L. monocytogenes* CC/ST (WGS) with CNS outcome to confirm/extend ST6 findings beyond the Netherlands.
2. **Host immunogenetics:** Targeted sequencing / immune-function studies in non-classical Listeria meningitis patients to identify inborn or acquired immune defects (e.g., in the IFN-γ/TNF axis).
3. **RCT of adjunctive therapy:** Evaluate gentamicin's incremental benefit and dexamethasone's harm/benefit specifically in confirmed Listeria meningitis.
4. **Neuroinvasion mechanism:** Use humanized E-cadherin mice + intravital imaging and the zebrafish model to dissect transcellular vs Trojan-horse BBB crossing and test anti-InlB/Met blockade.
5. **Biomarker development:** Assess CSF/serum markers and MRI hydrocephalus/parenchymal-lesion scoring as validated prognostic tools for early neurocritical escalation.

6. **Prevention impact:** Quantify the effect of WGS-based food-surveillance programs on invasive listeriosis incidence in risk groups.
